Probability of Default Model

Home Credit Default Risk

Abdul Malik

Overview

A **Probability of Default (PD) model** estimates the likelihood that a borrower will fail to repay a loan. It is a core component of retail credit risk management and a regulatory requirement under the Basel II/III Internal Ratings-Based (IRB) framework, where banks must assign a quantified PD to every credit exposure.

This project builds a PD model on the Home Credit Default Risk dataset — a publicly available dataset of 307,511 consumer loan applications with known repayment outcomes. The dataset reflects real-world challenges common in retail lending: class imbalance (≈8% default rate), high missingness in key predictors, and a mix of behavioural, bureau, and demographic signals.

Two models are built and compared:

- **WoE Scorecard (primary)** — A logistic regression model trained on Weight of Evidence (WoE) transformed features. This is the Basel IRB standard approach used by retail banks. Every feature is binned, assigned a WoE value, and transformed into scorecard points, producing a single integer score per applicant that can be fully decomposed and explained.
- **LightGBM (benchmark)** — A gradient boosted tree model trained on the same feature set. Included to quantify the performance cost of interpretability — how much Gini is sacrificed by choosing a transparent model over a black-box one.

The primary output is the scorecard. LightGBM serves as a reference point, not an alternative for deployment.

---

1. Dataset

Population	307,511 loan applications
Target	`TARGET` — 1 = defaulted, 0 = repaid
Overall default rate	8.07% (24,825 defaults)
Train set	246,008 rows — default rate 8.11% (19,941 defaults)
Out-of-time (OOT)	61,503 rows — default rate 7.94% (4,884 defaults)
Rate difference	0.17 pp — negligible, no evidence of temporal drift

Segment	Group	Default rate
Contract type	Cash loans	8.3%
	Revolving loans	5.5%
Gender	Male	10.1%
	Female	7.0%
Income type	Maternity leave	40.0%
	Unemployed	36.4%
	Working	9.6%
	State servant	5.8%
	Pensioner	5.4%
Education	Lower secondary	10.9%
	Secondary	8.9%
	Incomplete higher	8.5%
	Higher education	5.4%
	Academic degree	1.8%

Table 1: Employment status and education show the steepest risk gradients.

2. EDA Findings

Default Rate by Segment

External Scores

Feature	No default (mean)	Default (mean)	Gap
EXT_SOURCE_1	0.511	0.387	0.124
EXT_SOURCE_2	0.523	0.411	0.112
EXT_SOURCE_3	0.521	0.391	0.130

Table 2: All three external scores clearly separate defaulters. Higher score = lower risk.

Mean applicant age: 44.2 years (no default) vs. 40.8 years (default). Older applicants default less; the 3.4-year gap is meaningful at scale.

Key Missing Values

Feature	Missing rate	Note
EXT_SOURCE_1	56.4%	Missingness is itself a risk signal
OWN_CAR_AGE	66.0%	Missing $\Rightarrow$ no car owned
OCCUPATION_TYPE	31.3%	Self-reported, often blank
EXT_SOURCE_3	19.8%	Moderate missingness
EXT_SOURCE_2	0.2%	Nearly complete

Table 3: WoE binning assigns a dedicated bin to missing values; no imputation required.

3. Features Selected

25 features selected automatically by Information Value > 0.05. Raw columns superseded by engineered ones were excluded.

Group	Features	What they measure
External scores	EXT_SOURCE_1/2/3, ext_source_mean, ext_source_min	Third-party credit scores
Credit card	cc_utilisation, cc_avg_balance, cc_avg_payment	Card usage and payments
Instalment history	inst_late_rate, inst_late_count, inst_max_late, inst_avg_diff	Payment timeliness
Bureau	bureau_debt_ratio, bureau_avg_days	External credit burden
Affordability	credit_annuity_ratio, credit_goods_ratio, AMT_GOODS_PRICE	Loan sizing vs capacity
Employment & age	employed_years, employed_age_ratio, age_years	Tenure and age
Categorical	OCCUPATION_TYPE, ORGANIZATION_TYPE, NAME_INCOME_TYPE, NAME_EDUCATION_TYPE	Socioeconomic profile
Assets	OWN_CAR_AGE	Asset ownership

4. WoE Scorecard — Logistic Regression Coefficients

In this WoE encoding, negative coefficients indicate protective features (higher WoE $\Rightarrow$ lower predicted default). Three features carry small positive coefficients, likely reflecting non-monotone binning.

5. Scorecard Points

The score scale is centred at 600 (base odds 50:1, PDO = 20). Defaulters score approximately 27 points lower on average, enabling the setting of cut-off thresholds.

6. Model Performance

LightGBM — Top Features by Gain

7. Calibration — Hosmer-Lemeshow

The scorecard predicts actual default rates well across the full risk spectrum.

8. Population Stability Index

PSI = 0.0003 — Stable (threshold: < 0.10).

The score distribution is virtually identical between train and OOT. The model shows no signs of overfitting to the training population.

Feature	Coefficient	Direction
credit_annuity_ratio	−0.644	Higher annuity-to-credit ratio → lower risk
OWN_CAR_AGE	−0.606	Older car ownership → lower risk
EXT_SOURCE_2	−0.538	Higher score → lower risk
EXT_SOURCE_3	−0.535	Higher score → lower risk
credit_goods_ratio	−0.477	Loan closer to goods value → lower risk
AMT_GOODS_PRICE	−0.467	Higher goods price → lower risk
NAME_EDUCATION_TYPE	−0.464	Higher education → lower risk
cc_utilisation	−0.458	Lower utilisation → lower risk
inst_max_late	−0.455	Fewer max late days → lower risk
ORGANIZATION_TYPE	−0.445	Organization type gradient
EXT_SOURCE_1	−0.376	Higher score → lower risk
bureau_debt_ratio	−0.349	Lower debt-to-credit → lower risk
OCCUPATION_TYPE	−0.333	Occupation type gradient
employed_age_ratio	−0.234	Longer employment relative to age → lower risk
employed_years	−0.225	Longer tenure → lower risk
ext_source_mean	−0.195	Higher composite score → lower risk
NAME_INCOME_TYPE	−0.176	Income type gradient
inst_late_rate	−0.158	Lower late payment rate → lower risk
inst_avg_diff	−0.152	Smaller payment shortfall → lower risk
cc_avg_balance	−0.137	Lower card balance → lower risk
bureau_avg_days	−0.101	More recent bureau entries → lower risk
ext_source_min	−0.093	Higher worst external score → lower risk
age_years	+0.055	*Positive — non-monotone WoE bins*
inst_late_count	+0.077	*Positive — non-monotone WoE bins*
cc_avg_payment	+0.098	*Positive — check WoE bins*

9. Rank-Ordering

10. Scorecard vs. LightGBM — The Trade-off

A bank accepts this 2.9-point Gini loss because:

- Basel II/III IRB requires that every credit decision be explainable to the applicant.
- Model risk teams require auditability — why did this score change between two dates?
- WoE bins and scorecard points make stress-testing and sensitivity analysis straightforward.
- LightGBM feature importance describes global behaviour only; it cannot explain a single customer's score.

	No default	Default
Mean score	570.3	543.8
Score range (OOT)	452 – 655	452 – 655
Separation	**26.5 points**	

Model	AUC	Gini	KS	Benchmark
WoE Scorecard	0.758	0.516	0.391	Gini > 0.50 ✓
LightGBM	0.773	0.545	0.409	Gini > 0.50 ✓
Gap	0.015	0.029	0.018	—

Table 4: Both models exceed the "Good" threshold for retail unsecured lending (Gini > 0.50). The Gini gap of 2.9 points is small.

11. Known Limitations

Models trained on Home Credit Default Risk dataset. Validation performed on an out-of-time hold-out set. All metrics reported on OOT only.

Feature	Gain	Note
ext_source_mean	112,813	Dominant — ≈5× the next feature
ORGANIZATION_TYPE	23,551	Employer sector
credit_annuity_ratio	19,279	Loan affordability
credit_goods_ratio	10,301	Loan-to-goods ratio
EXT_SOURCE_3	9,001	External score
AMT_GOODS_PRICE	8,628	Goods value
bureau_debt_ratio	8,427	External credit burden
inst_max_late	8,114	Worst payment delay
cc_utilisation	8,068	Card utilisation
ext_source_min	7,580	Worst external score

Table 5: External scores dominate. The composite ext_source_mean alone contributes more information gain than the next four features combined.

Decile	N	Observed rate	Expected PD	Assessment
1 (lowest PD)	6,151	1.5%	1.3%	✓
2	6,150	1.7%	2.2%	✓
3	6,150	2.8%	3.0%	✓
4	6,150	3.0%	3.8%	✓slight under
5	6,151	4.5%	4.8%	✓
6	6,150	5.9%	6.1%	✓
7	6,150	8.3%	7.9%	✓
8	6,150	10.7%	10.4%	✓
9	6,150	15.1%	14.6%	✓
10 (highest PD)	6,151	25.9%	26.6%	✓slight over

Table 6: Observed and expected default rates track closely across all deciles.

Decile	Count	Default rate	Avg PD	Cum. defaults captured
1 (riskiest)	6,151	**25.9%**	26.6%	32.6%
2	6,150	15.1%	14.6%	51.6%
3	6,150	10.7%	10.4%	65.1%
4	6,150	8.3%	7.9%	75.6%
5	6,150	5.9%	6.1%	83.0%
6	6,151	4.5%	4.8%	88.7%
7	6,150	3.0%	3.8%	92.5%
8	6,150	2.8%	3.0%	96.0%
9	6,150	1.7%	2.2%	98.1%
10 (safest)	6,151	**1.5%**	1.3%	100%

Table 7: Decile 1 default rate is 17× that of decile 10. The top two deciles capture 51.6% of all defaults while representing 20% of applicants.

	WoE Scorecard	**LightGBM**
Gini	0.516	0.545
Interpretable	Yes — WoE bins and points per feature	Feature importance only
Per-applicant explanation	Full decomposition	Not available
Regulatory fit	Yes (Basel IRB)	No
Audit trail	Full	Not available
Cost of interpretability	**2.9 Gini points**	

Limitation	**Detail**
`bureau_balance` excluded	27M rows, 3 columns — skipped for efficiency; may contain additional delinquency signal
`POS_CASH_balance` excluded	Similar signal to instalments; skipped
EXT_SOURCE opacity	The strongest predictors are external scores of unknown construction
OOT split is approximate	Sorted by customer ID as a proxy for time — not a guaranteed temporal split
25 features vs target of 15–20	All pass IV $>$ 0.05; a tighter threshold would reduce the feature set
SHAP not available	Replaced by LightGBM built-in importance due to a library version conflict